Mandatory Physical Calibration Guide

DO NOT PRINT THE FULL ENCLOSURE YET. PRINT AND COMPLETE THE CALIBRATION PARTS FIRST.

Use the same material, nozzle, layer height, wall count, extrusion settings, bed preparation, and chamber temperature that you will use for the enclosure.

Print first

Print these exact files from `CALIBRATION_PARTS/`:

1. `coupon_official_interface.3mf`
2. `coupon_heat_set_insert.3mf`
3. `coupon_active_driver.3mf`
4. `coupon_passive_radiator.3mf`
5. `coupon_gasket_base.3mf`
6. `coupon_gasket_cap.3mf`
7. `coupon_cable_passage.3mf`
8. `cable_gland.3mf` in TPU 95A

ASA baseline: 0.4 mm nozzle, 0.20 mm layers, five walls, six top/bottom layers, 35% gyroid, 100-110 C bed, 250-260 C nozzle, enclosure closed, no supports, 5 mm brim. PETG alternative: 235-250 C nozzle, 75-85 C bed, three-hour dry cycle if stringing is present.

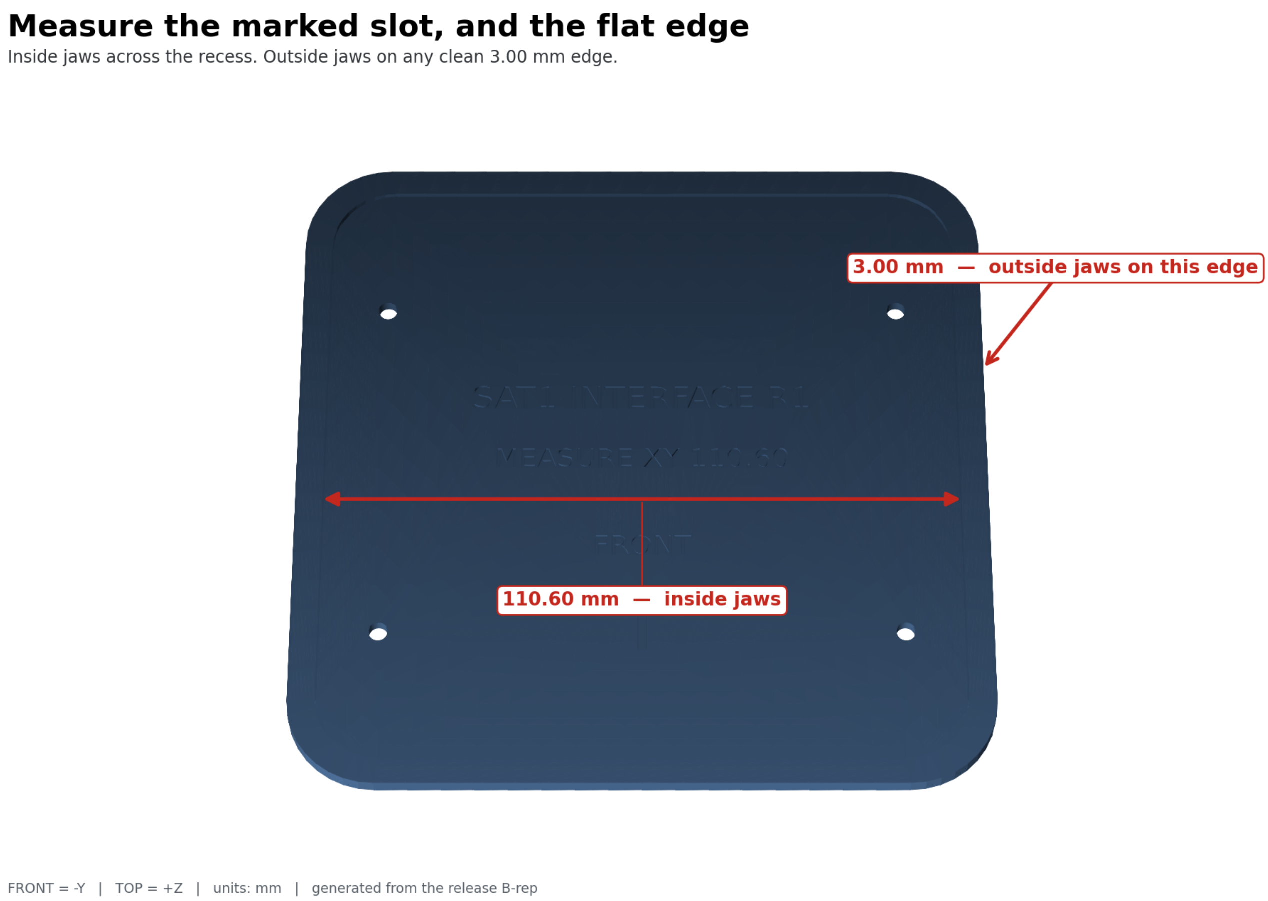
Measure and enter values

Use the engraved labels and `IMAGES/calibration_*.png`.

Check	Where/how	Nominal	Pass	Input key
XY scale	Inside jaws across `MEASURE XY 110.60` recess, three heights	110.60 mm	110.40-110.80 mm after correction	`xy_scale_correction_fraction = 110.60 / measured - 1`
Z scale	Outside jaws across clean 3.00 mm coupon edge, four corners	3.00 mm	2.90-3.10 mm	`z_scale_correction_fraction = 3.00 / measured - 1`
M3 clearance	Try a clean ISO M3 screw in labeled 3.4/3.5/3.6 holes	3.4 mm	smallest hole that falls through without force	chosen diameter minus 3.4
Insert bore	Install identical inserts in 4.0/4.1/4.2/4.3 blind bores	4.2 mm	square, flush, no crack/spin at 0.35 N m	chosen diameter minus 4.2
Driver fit	Seat the purchased ND91-4 in the labeled coupon	catalog interface	drops in by hand, <=0.30 mm radial play, flange lies flat	cutout correction and measured flange thickness
Radiator fit	Seat one passive radiator in the labeled coupon	catalog interface	drops in by hand, <=0.30 mm radial play, flange lies flat	cutout correction and measured flange thickness
Gasket	Tighten cap on a strip of the actual sheet until both stops contact	2.00 to 1.50 mm	15%-45% compression; no open light path	sheet thickness and compressed-thickness offset
Cable gland	Fit actual two 22 AWG conductors and gland in cable coupon	8.0 mm passage	moderate finger force; gland cannot rotate or lift	cable-passage offset

Do not use caliper tips to measure a 3-4 mm hole; the screw and insert are the functional gauges. Enter only values you measured.

CAD measurement illustrations



Official-interface XY and Z measurement

Use the screw and the insert as gauges

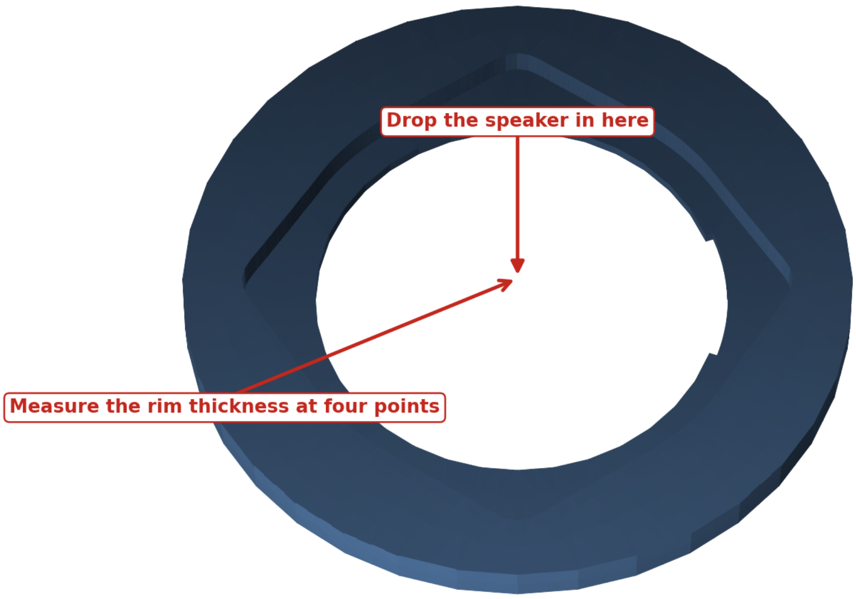
Do not measure these holes with caliper tips. The hardware is the gauge.

FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

M3 screw and insert functional gauges

Check your speaker drops in

It must drop in by hand and sit flat. Never force it.

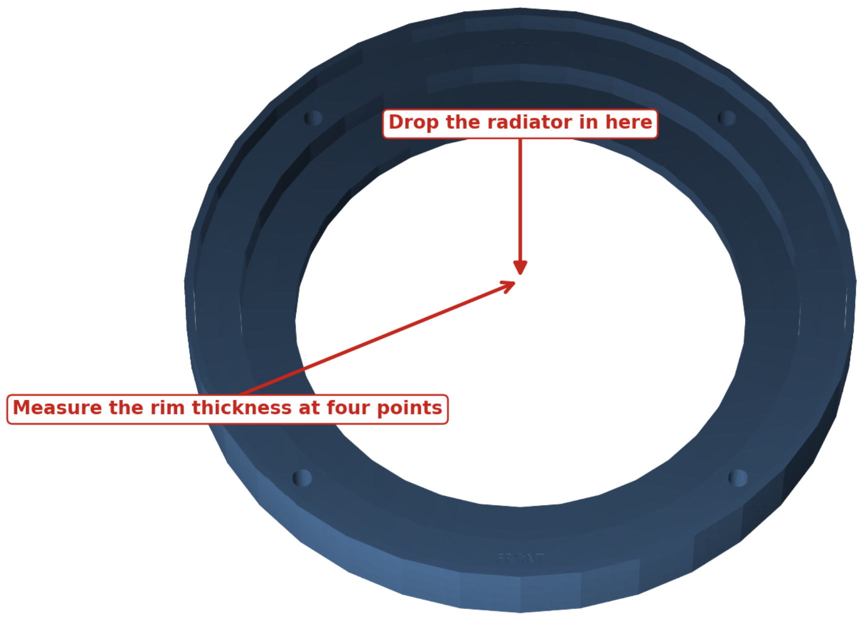


FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

Active-driver coupon fit check

Check your radiator drops in

It must drop in by hand and sit flat. Never force it.

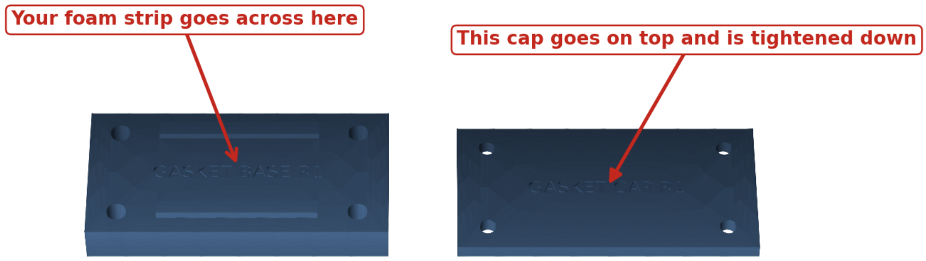


FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

Passive-radiator coupon fit check

Squash a strip of your foam between these two

Tighten until the two hard stops touch, then measure the gap that is left.

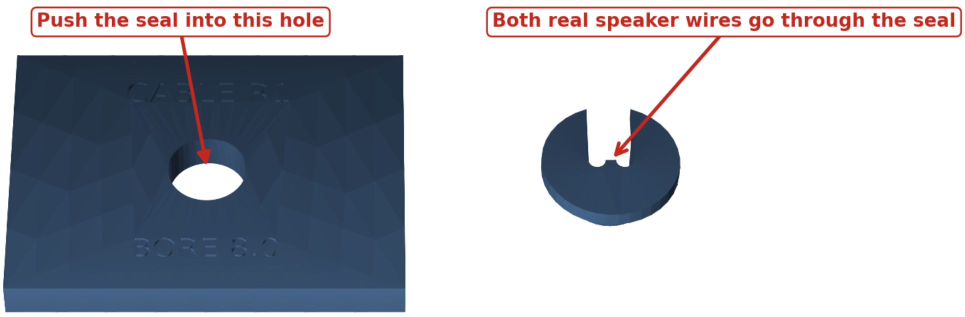


FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

Gasket compression coupon stack

Push the seal and both wires through the hole

It should take firm finger pressure, then stay put and refuse to twist.



FRONT = -Y | TOP = +Z | units: mm | generated from the release B-rep

Cable passage and TPU gland check

Edit one file

Copy `CALIBRATION_INPUT_TEMPLATE.yaml` to `config/physical_calibration.yaml`, or run:

```
python scripts/calibrate.py
```

The file exposes only user-facing corrections. Safe limits reject implausible values before any full part is built.

Regenerate

```
make calibrated-release
```

Success ends with all validation, documentation, mutation, and package checks passing. Output is in `release/Satellite1-Ultra-RC1/`.

Example successful finish:

```
documentation PASS; 9 guides, 7 PDFs
release/Satellite1-Ultra-RC1 (all required files present)
```

Reprint every coupon affected by a nonzero correction. You are cleared for the full enclosure only when all eight checks above pass on the corrected coupons.

Common failures

- Warped official coupon: improve enclosure temperature, clean the bed, add the brim, and do not compensate a warped part.
- Every hole undersized: check flow and elephant-foot compensation before adding a large CAD offset.
- Insert cracks the boss: choose a larger coupon bore or reduce iron dwell; do not increase torque.
- Component will not sit flat: remove only print strings. Do not sand a sealing land; correct the cutout and reprint the coupon.
- Gland leaks or spins: verify conductor OD <=1.8 mm, dry TPU, then correct and reprint both the coupon and gland.

Source commit e973d7a60484. No physical validation has been performed.